Ministerul Educației și Cercetării al Republicii Moldova
Universitatea Tehnică a Moldovei
Facultatea Calculatoare, Informatică și Microelectronică

Raport
la
Lucrarea de laborator Nr. 7

Disciplina: Tehnici de programare

Tema: ” GitHub”

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A verificat : asistent universitar Reițman Patricia

Chișinău – 2026

SARCINA LUCRĂRII

Lab A

A1. Contul GitHub

A2. Git pe calculator: verifici și te prezinți

A3. Primul depozit, creat pe GitHub

A4. Îl aduci pe calculator: git clone

A5. Prima ta fotografie: status, add, commit

A6. O trimiți pe GitHub: git push

A7. O schimbare făcută pe GitHub, adusă înapoi: git pull

A8. Ce NU urcă pe GitHub: .gitignore

A9. Citești istoria și schimbările: log și diff

A10. Al doilea depozit, pornit de pe calculator

A11. Verificarea finală

Lab B

B1. Pregătire, în depozitul exercitiu-git

B2. Metoda, la fiecare eroare

B3. Greșelile care nu lasă programul să pornească

B4. Greșelile de nume, de tip și de date

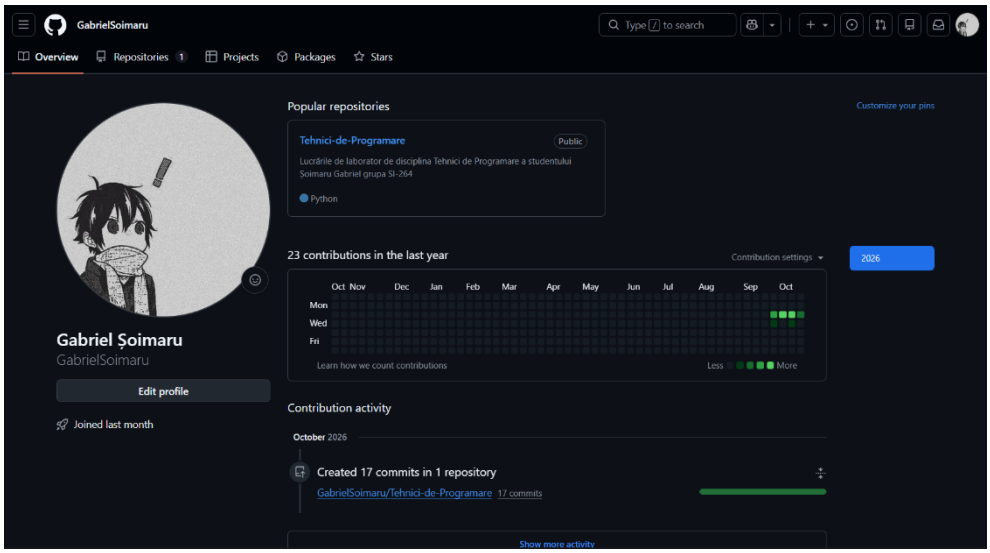
B5. Greșelile tăcute

B6. Verificarea finală și push

EFECTUAREA LUCRĂRII

Lucrarea A

A1



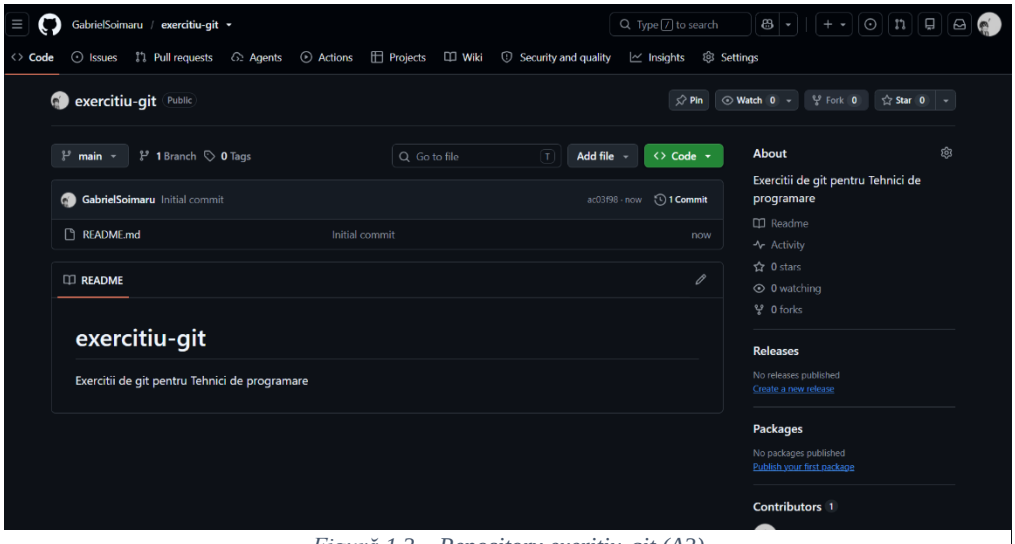
Figură 1.1 – Contul GitHub (A1)

A2

```
C:\Windows\system32>git config --global --list
core.editor="C:\Users\Gabriel\AppData\Local\Programs\Microsoft VS Code\bin\code" --wait
user.name=Gabriel Șoimaru
user.email=gabriel.soimaru@stud.fcim.utm.md
init.defaultbranch=main
pull.rebase=false
```

Figură 1.2 – Git pe calculator (A2)

A3



Figură 1.3 – Repository exeritiu-git (A3)

A4

```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub>git clone https://github.com/GabrielSoimaru/exercitiu-git.git
Cloning into 'exercitiu-git'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub>cd exercitiu-git

C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>dir
 Volume in drive C has no label.
 Volume Serial Number is 6E36-00B6

 Directory of C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git

10/24/2026  01:47 PM    <DIR>          .
10/24/2026  01:47 PM    <DIR>          ..
10/24/2026  01:47 PM                64 README.md
               1 File(s)             64 bytes
               2 Dir(s)  13,633,961,984 bytes free

C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>
```

Figură 1.4 – Clonarea Git (A4)

A5

```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>python salut.py
Salut, GitHub!
```

Figură 1.5.1 – Rularea python salut.py (A5)

```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        salut.py

nothing added to commit but untracked files present (use "git add" to track)
```

Figură 1.5.2 – Rularea git status (A5)

```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git add salut.py

C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   salut.py
```

Figură 1.5.3 – Rularea git add salut.py si git status (A5)

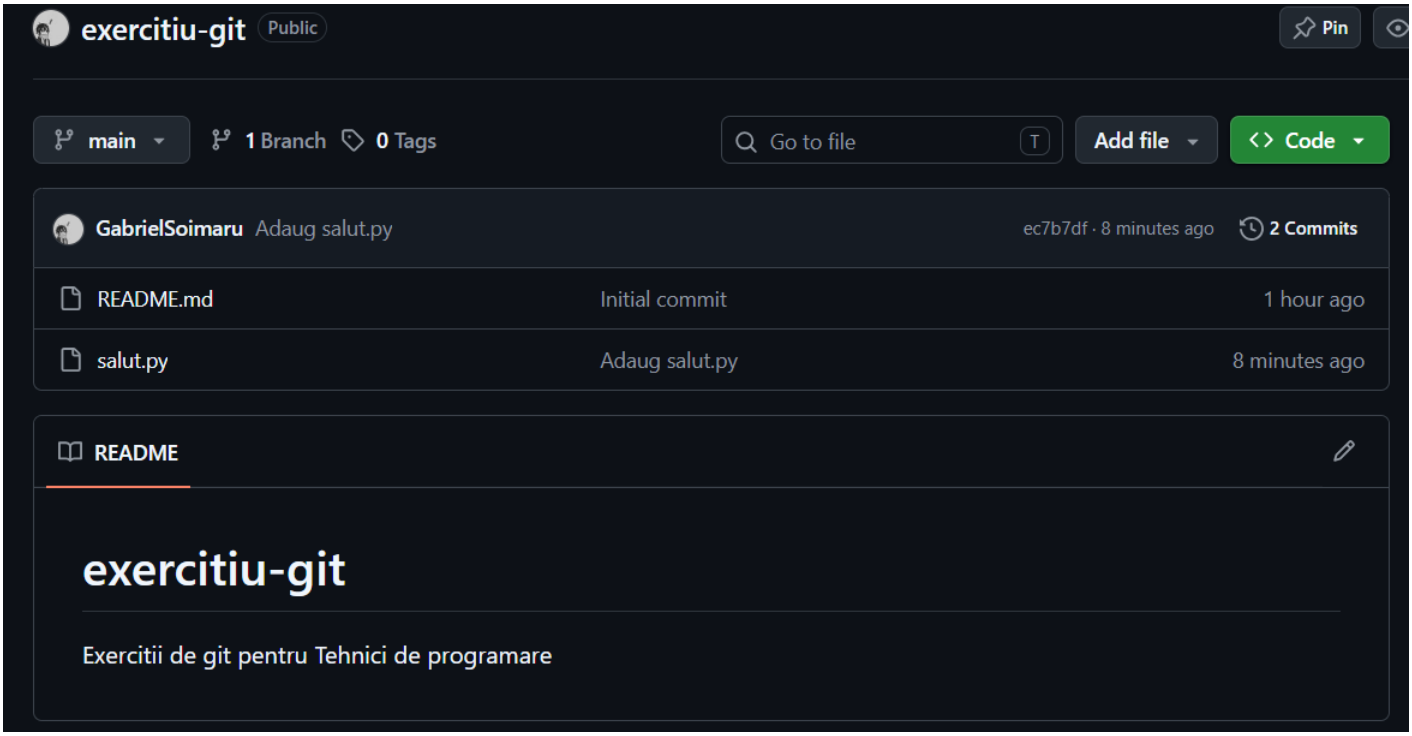
```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git commit -m "Adaug salut.py"
[main ec7b7df] Adaug salut.py
 1 file changed, 1 insertion(+)
 create mode 100644 salut.py
```

Figură 1.5.4 – Rularea git commit -m "Adaug salut.py" (A5)

A6

```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git push
info: please complete authentication in your browser...
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 316 bytes | 316.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/GabrielSoimaru/exercitiu-git.git
   ac03f98..ec7b7df  main -> main
```

Figură 1.6.1 – Rularea git push (A6)



Figură 1.6.2 – Dovada pe GitHub (A6)

A7

```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1008 bytes | 84.00 KiB/s, done.
From https://github.com/GabrielSoimaru/exercitiu-git
   ec7b7df..c2700f5  main       -> origin/main
Updating ec7b7df..c2700f5
Fast-forward
 README.md | 1 +
 1 file changed, 1 insertion(+)
```

Figură 1.7.1 – Rularea git pull (A7)

```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>type README.md
# exercitiu-git
Exercitii de git pentru Tehnici de programare
Aici exersez git.
```

Figură 1.7.2 – Rularea type README.md (A7)

A8

```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git status --short
?? .gitignore
```

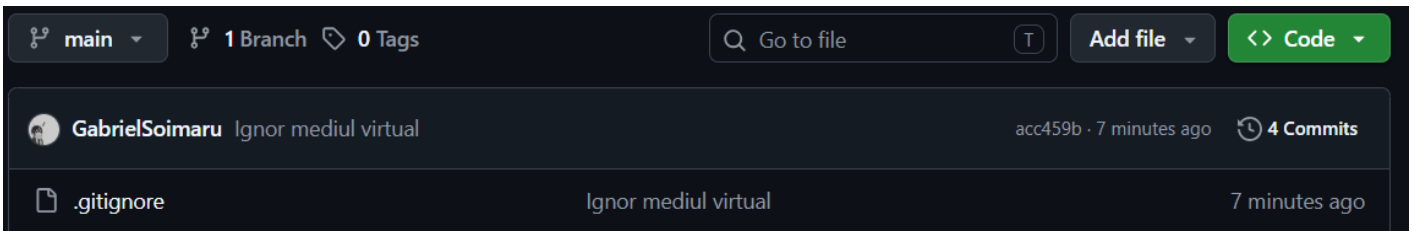
Figură 1.8.1 – Crearea .gitignore si rularea git status --short (A8)

```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git add .gitignore

C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git commit -m "Ignor mediul virtual"
[main acc459b] Ignor mediul virtual
 1 file changed, 2 insertions(+)
 create mode 100644 .gitignore

C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 351 bytes | 351.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/GabrielSoimaru/exercitiu-git.git
   c2700f5..acc459b  main -> main
```

Figură 1.8.2 – Adugarea .gitignore, comiterea lui și push pe GitHub (A8)



Figură 1.8.3 – Dovada pe GitHub (A8)

A9

```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git log --oneline
acc459b (HEAD -> main, origin/main, origin/HEAD) Ignor mediul virtual
c2700f5 Add a line about practicing Git
ec7b7df Adaug salut.py
ac03f98 Initial commit
```

Figură 1.9.1 – Rularera git log --oneline (A9)

```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git diff
diff --git a/salut.py b/salut.py
index c260f15..171acc9 100644
--- a/salut.py
+++ b/salut.py
@@ -1 +1,2 @@
-print("Salut, GitHub!")
\ No newline at end of file
+print("Salut, GitHub!")
+print("al doilea")
```

Figură 1.9.2 – Rularera git diff (A9)

```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git restore salut.py

C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Figură 1.9.3 – Rularea git restore salut.py și git status (A9)

A10

```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub>mkdir exercitiu-2

C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub>cd exercitiu-2

C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-2>git init
Initialized empty Git repository in C:/Users/Gabriel/Desktop/Gabi/Uni/Tehnici de programare/LabTP7/Lab_GitHub/exercitiu-2/.git/

C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-2>code README.md

C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-2>git add README.md

C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-2>git commit -m "Primul commit"
[main (root-commit) 3a49b34] Primul commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
```

Figură 1.10.1 – Crearea noului repository și fișierul README.md (A10)

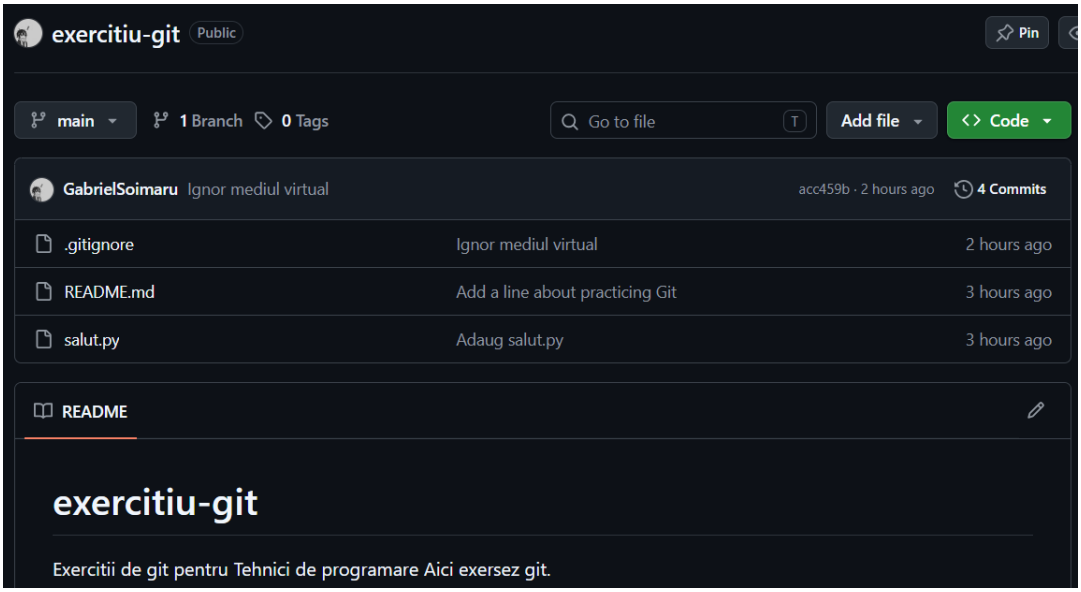
```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-2>git remote add origin https://github.com/GabrielSoimaru/exercitiu-2.git

C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-2>git branch -M main

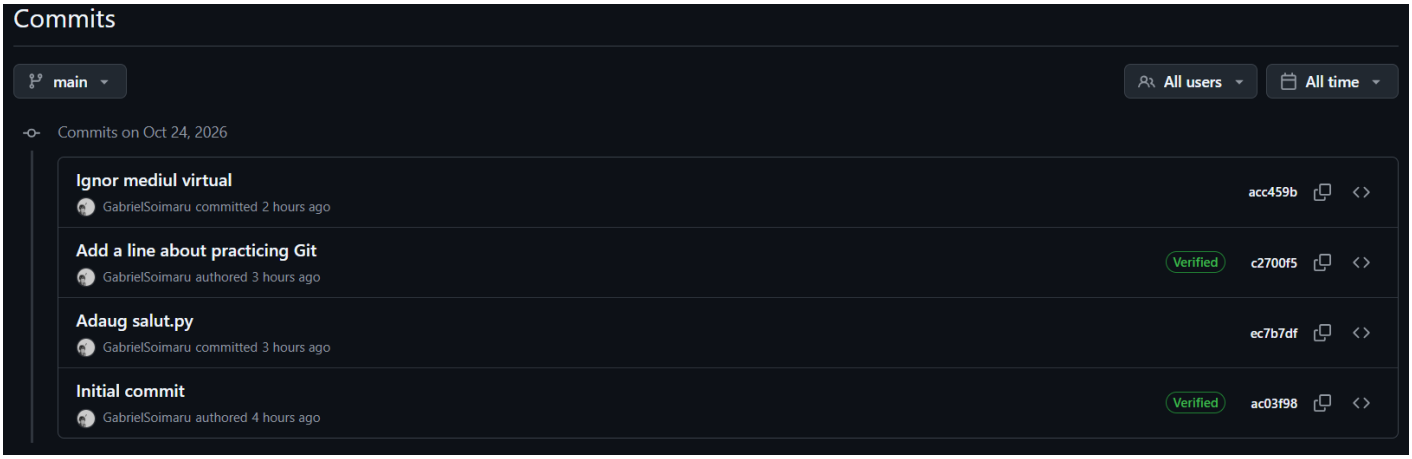
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-2>git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 255 bytes | 255.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/GabrielSoimaru/exercitiu-2.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Figură 1.10.2 – Push la repository exercitiu-2 (A10)

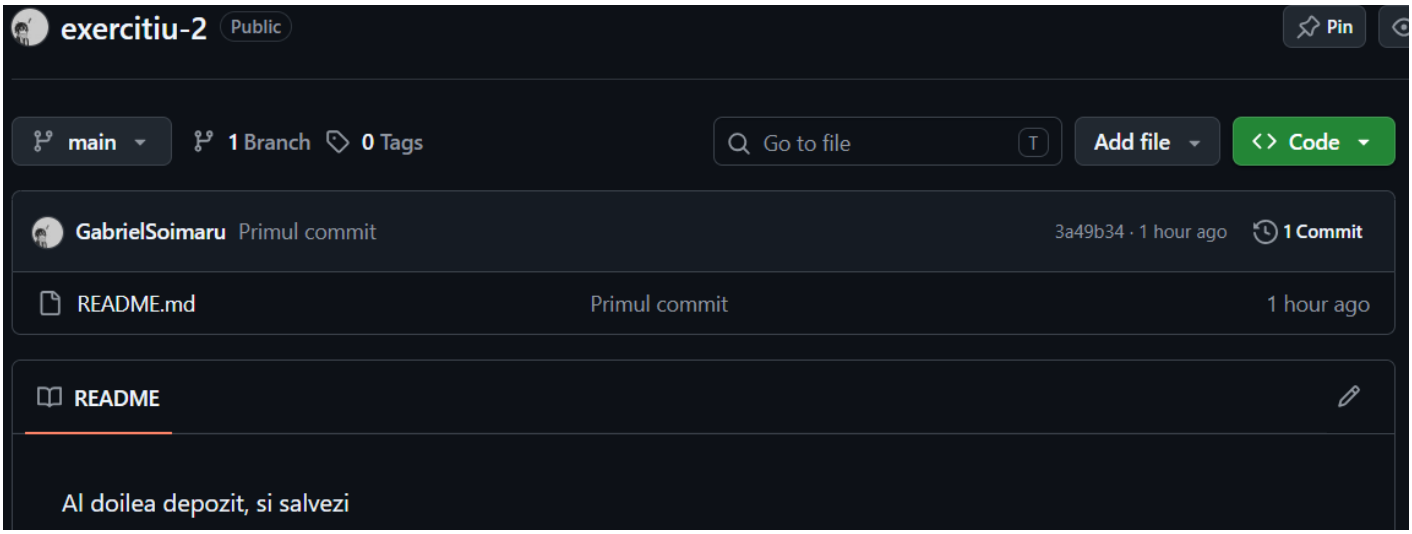
A11



Figură 1.11.1 – Repository exercitiu-git (A11)



Figură 1.11.2 – Istoria de commits în repository exercitiu-git (A11)



Figură 1.11.3 – Repository exercitiu-2 (A11)

Lucrarea B

B1

```
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>mkdir depanare

C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git add depanare
warning: in the working copy of 'depanare/inventar.py', LF will be replaced by CRLF the next time Git touches it

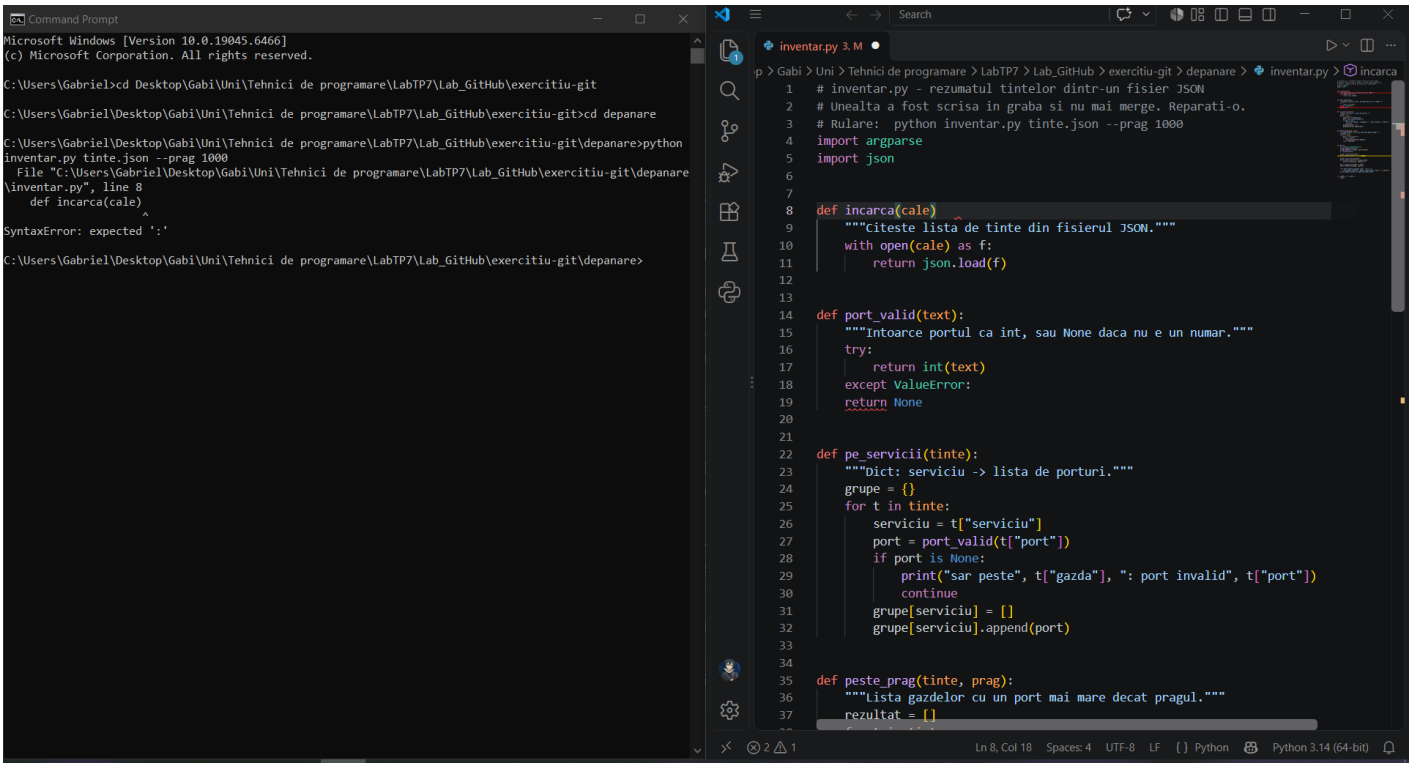
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>git commit -m "Pornire: inventar.py stricat"
[main 228630d] Pornire: inventar.py stricat
 2 files changed, 75 insertions(+)
 create mode 100644 depanare/inventar.py
 create mode 100644 depanare/tinte.json

C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git>cd depanare

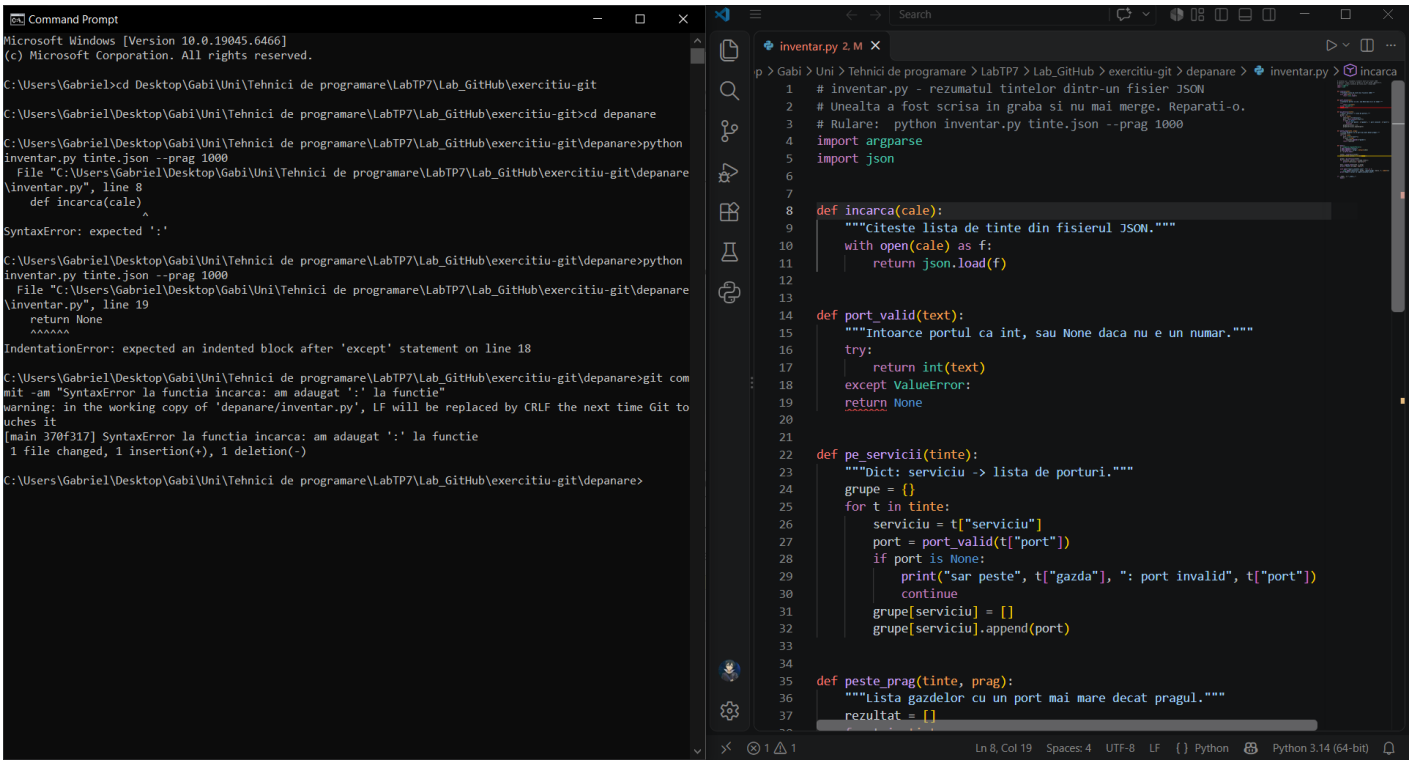
C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git\depanare>python inventar.py tinte.json --prag 1000
  File "C:\Users\Gabriel\Desktop\Gabi\Uni\Tehnici de programare\LabTP7\Lab_GitHub\exercitiu-git\depanare\inventar.py", line 8
    def incarca(cale)
                    ^
SyntaxError: expected ':'
```

Figură 2.1 – Crearea folder-ului depanare, copierea invetar.py și tinte.json (B1)

B2

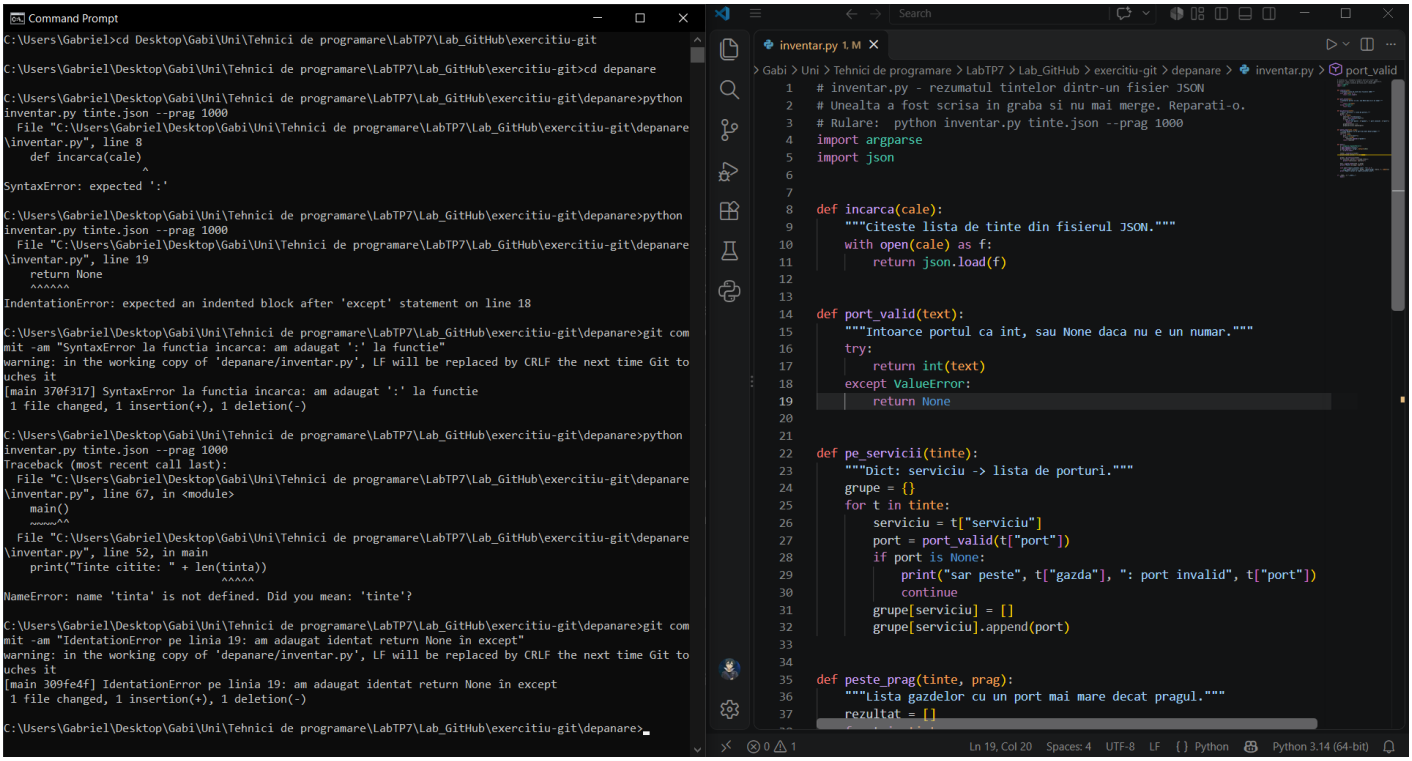


Figură 2.2.1 – Am citit și găsit eroarea (B2)



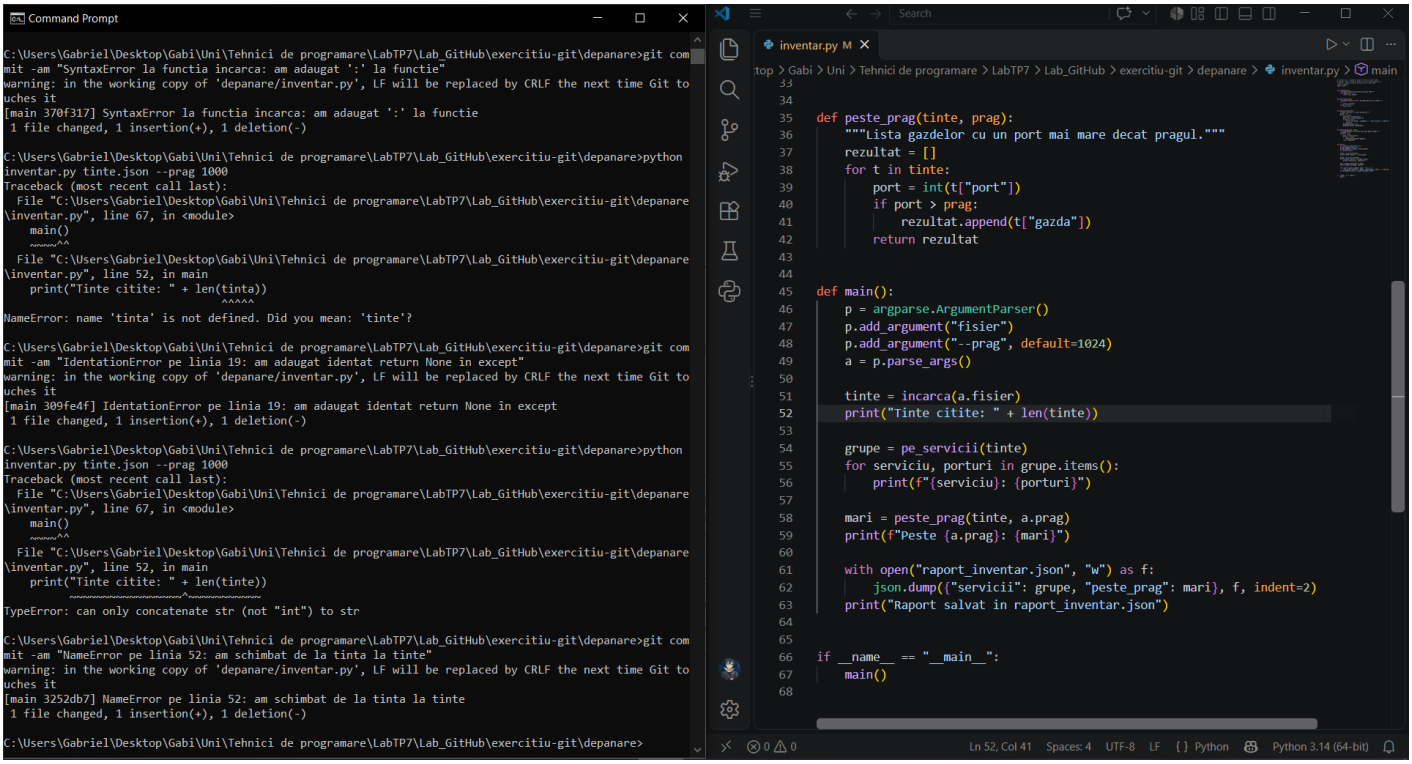
Figură 2.2.2 – Am reparat eroarea, am rulat încă odată și am făcut commit -am (B2)

B3

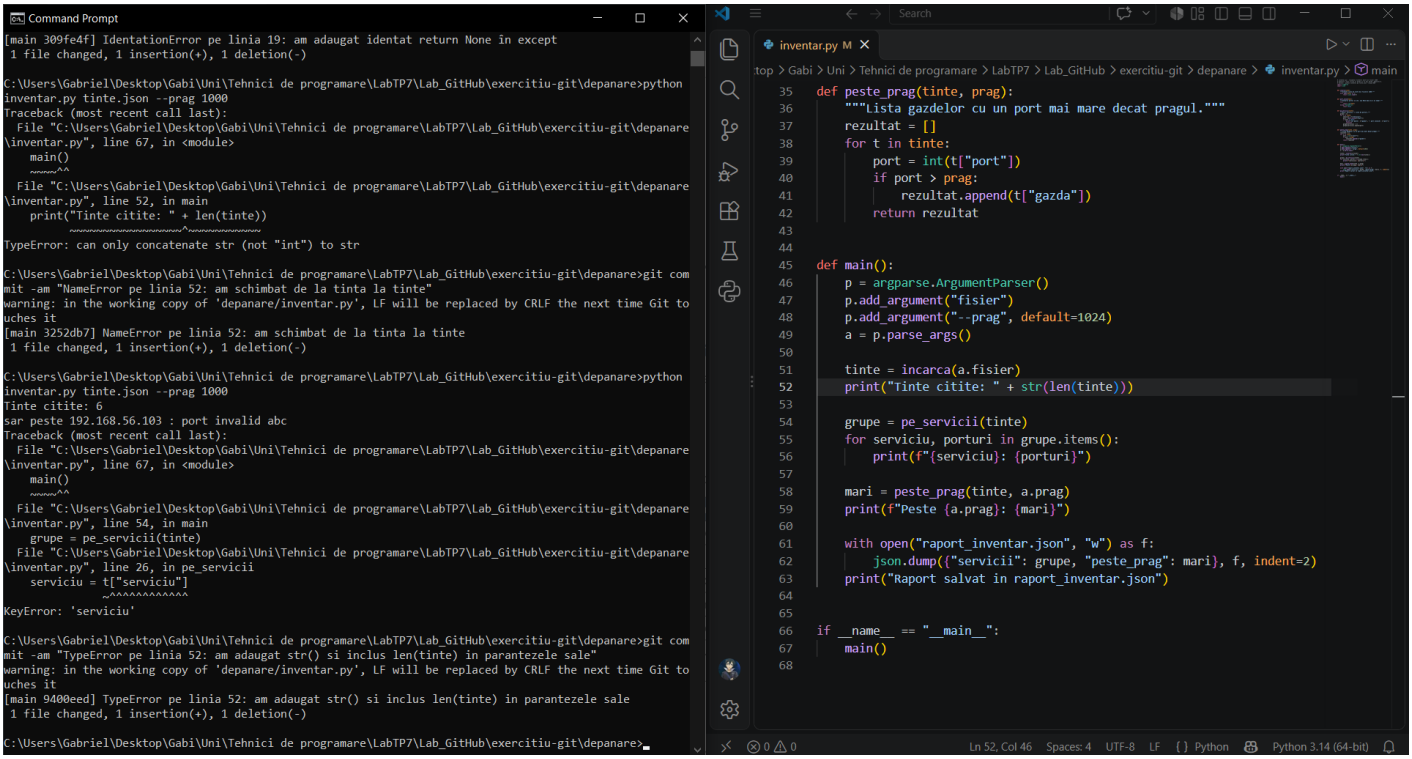


Figură 2.3 – Am reparat eroarea de tip IdentationError și am urmat pașii stabiliți în B2 (B3)

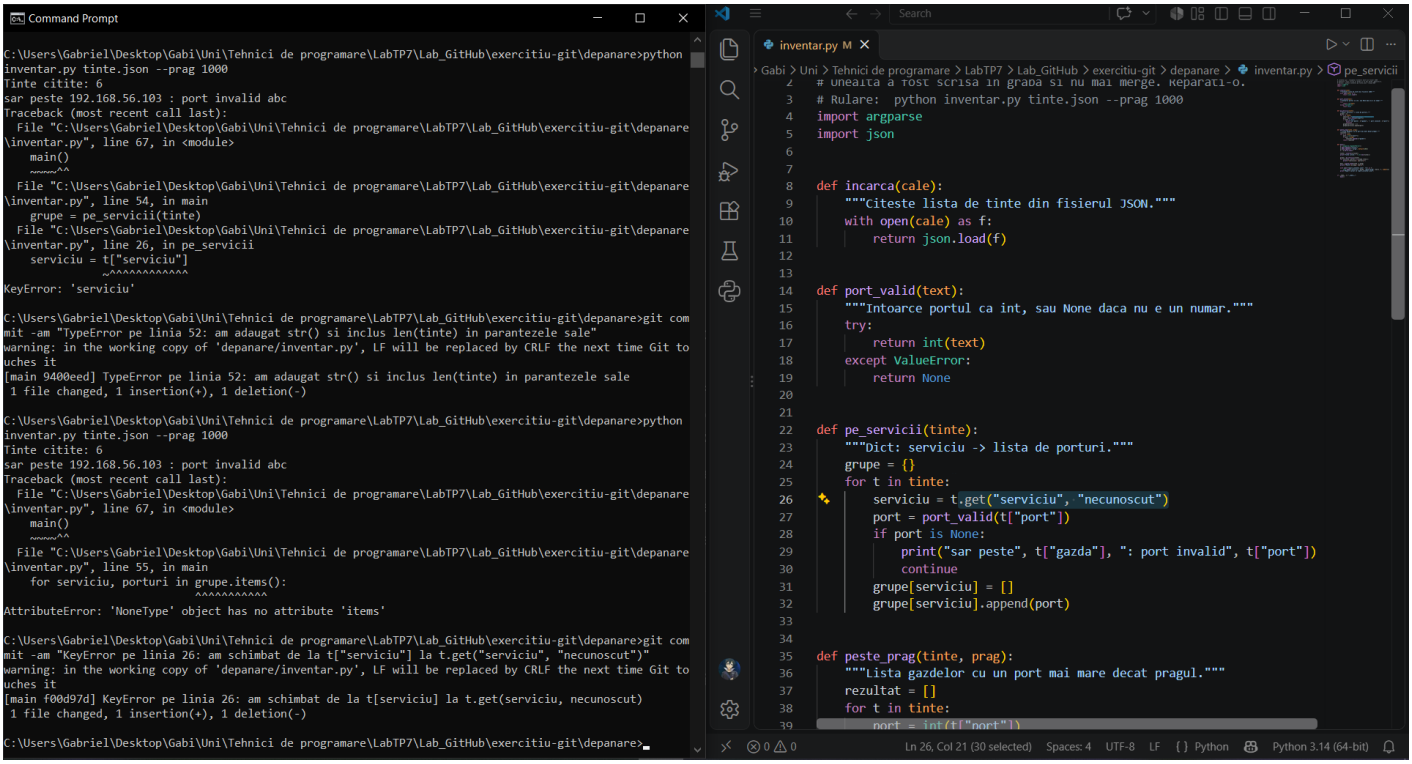
B4



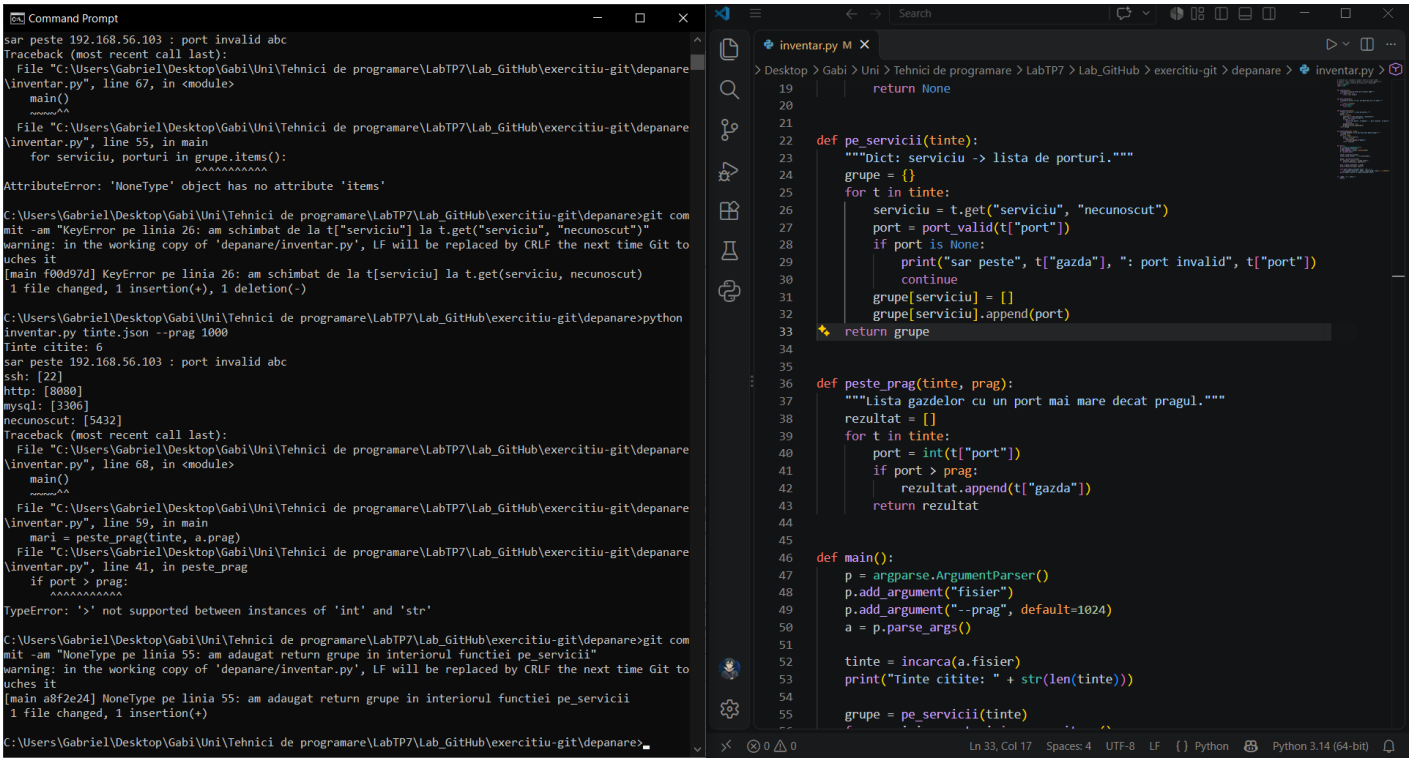
Figură 2.4.1 – Am reparat o eroare de tip NameError și am urmat algoritmul din B2 (B4)



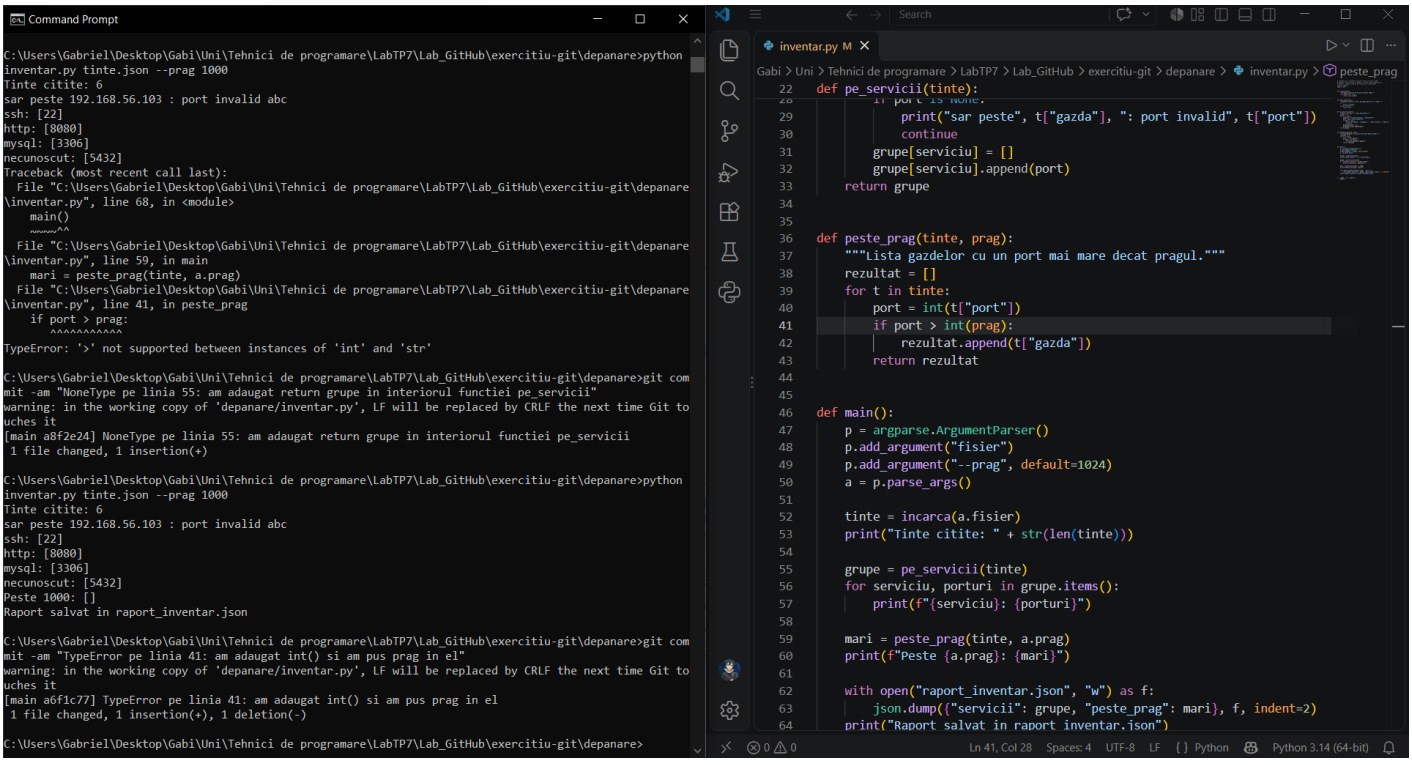
Figură 2.4.2 – Am reparat o eroare de tip TypeError și am urmat algoritmul din B2 (B4)



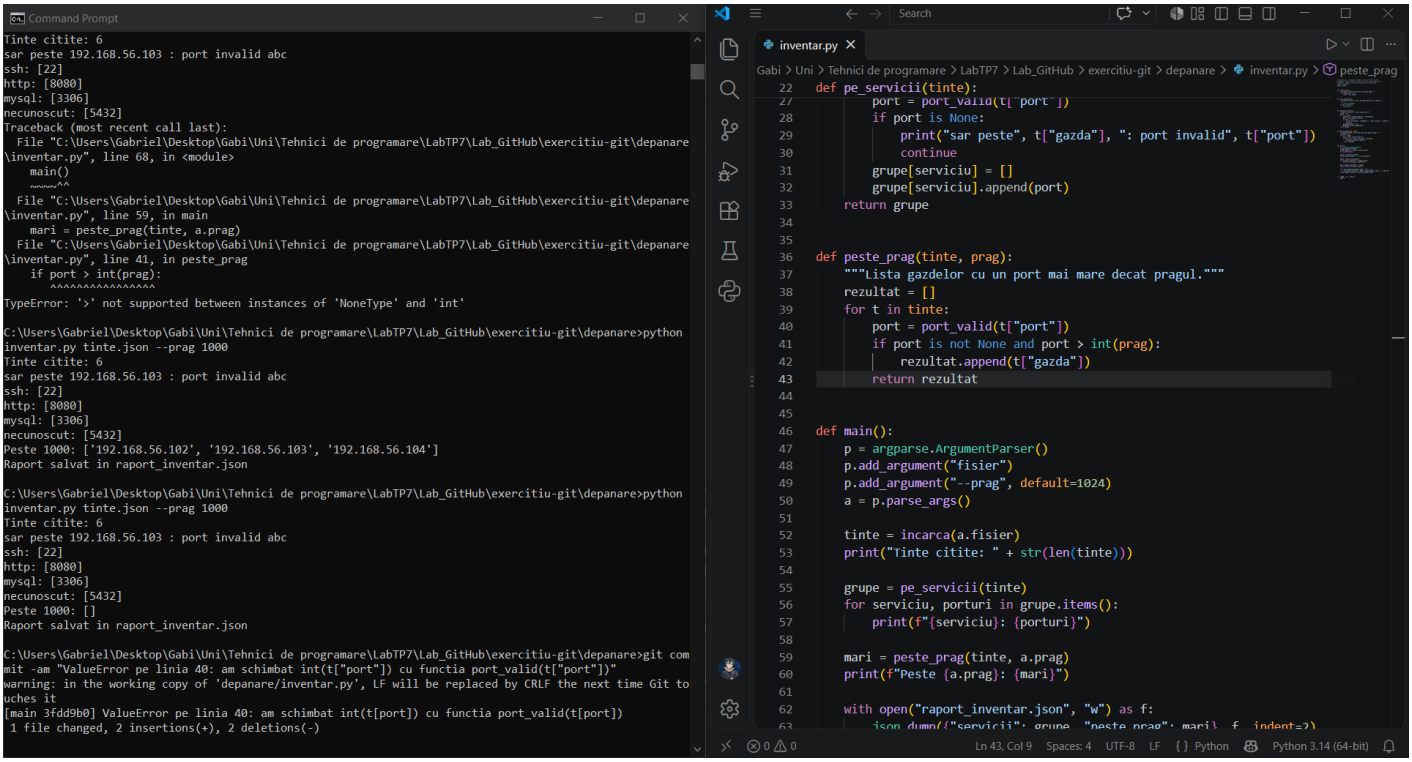
Figură 2.4.3 – Am reparat o eroare de tip KeyError și am urmat algoritmul din B2 (B4)



Figură 2.4.4 – Am reparat o eroare de tip NoneType și am urmat algoritmul din B2 (B4)

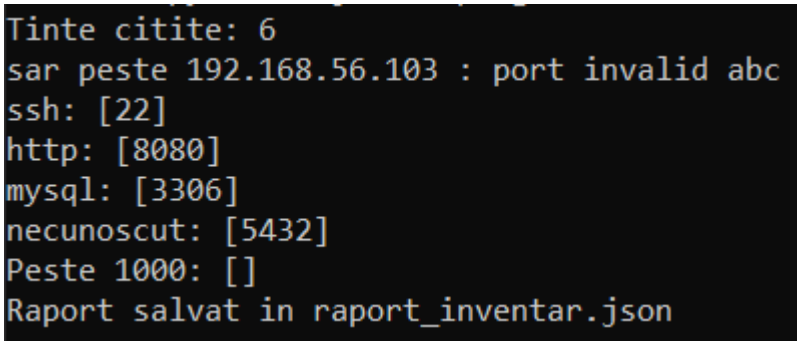


Figură 2.4.5 – Am reparat o eroare de tip TypeError și am urmat algoritmul din B2 (B4)

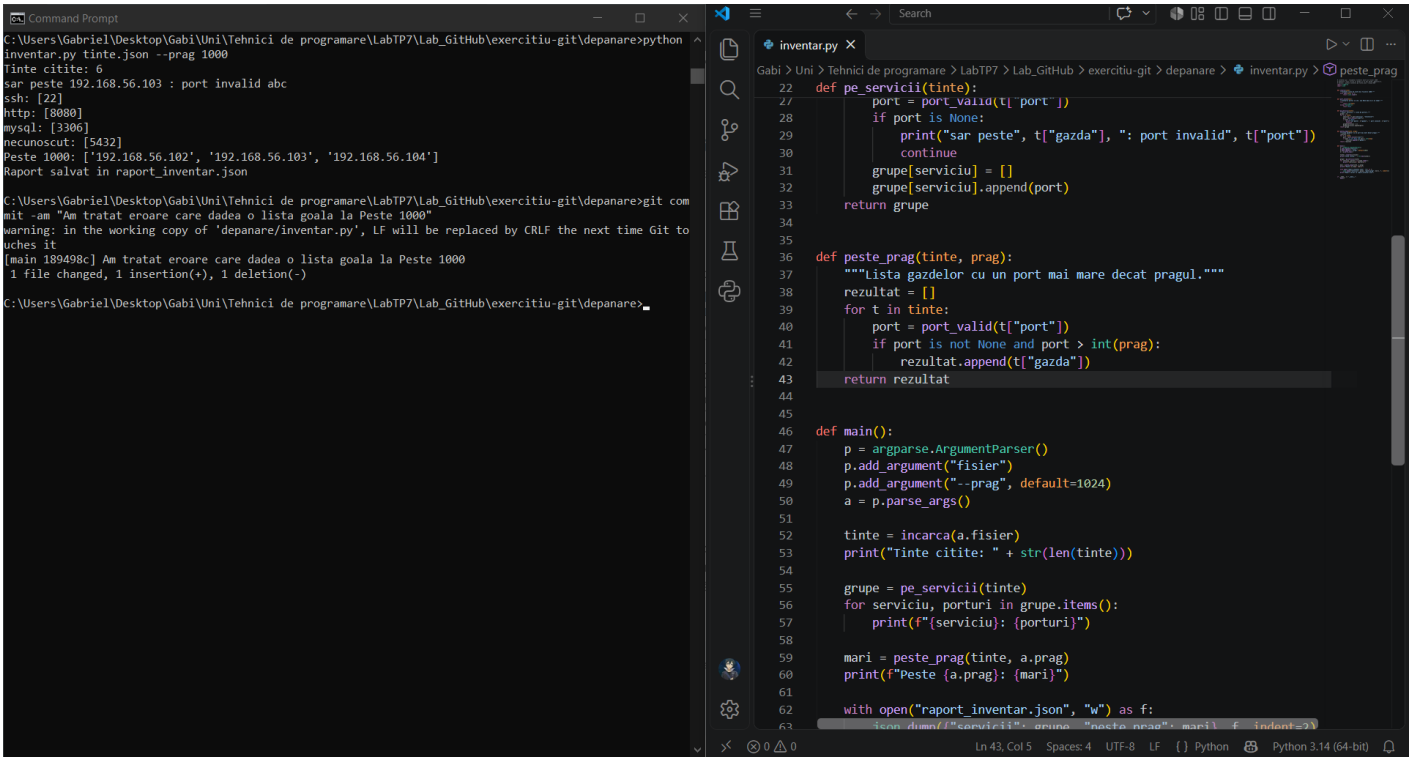


Figură 2.4.6 – Am reparat o eroare de tip ValueError și una de tip TypeError și am urmat algoritmul din B2 (B4)

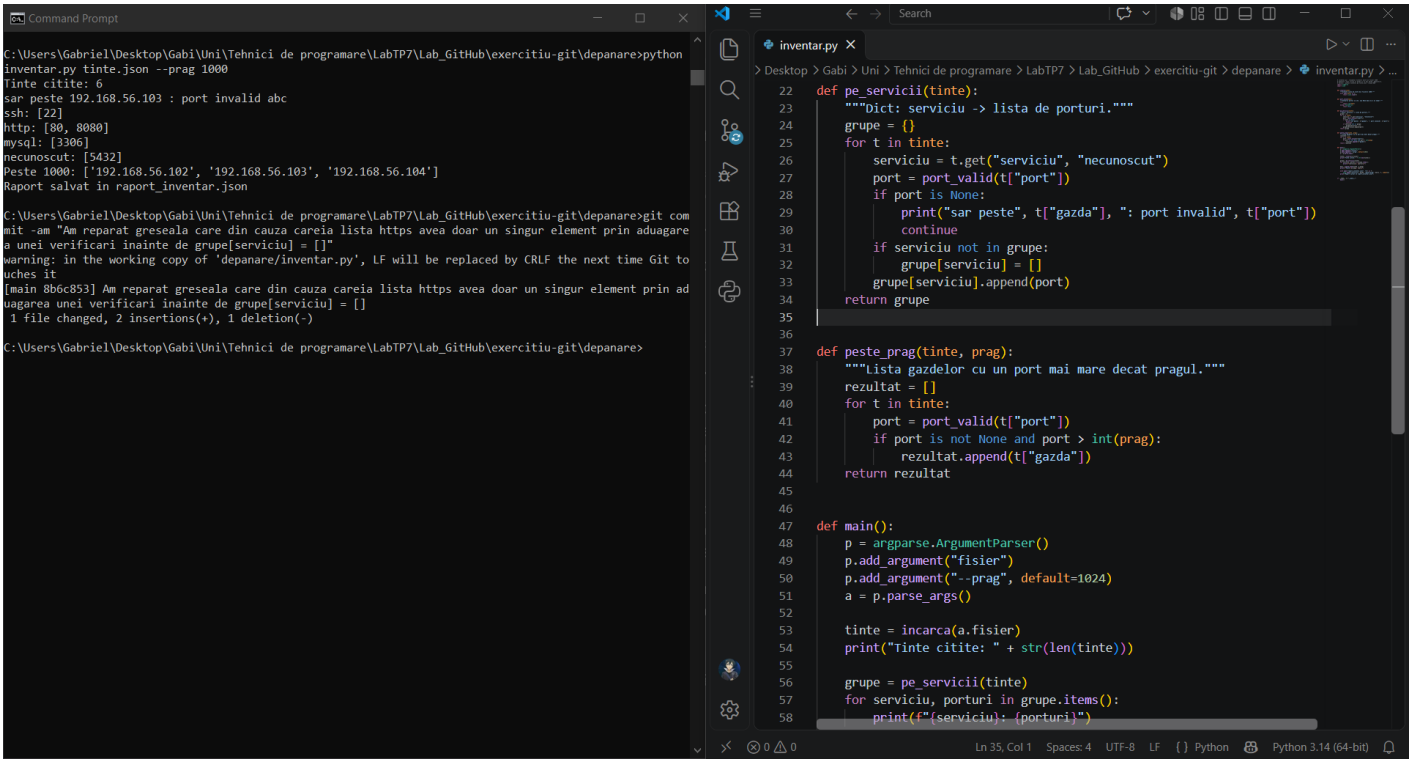
B5



Figură 2.5.1 – Rezultatul final greșit (B5)

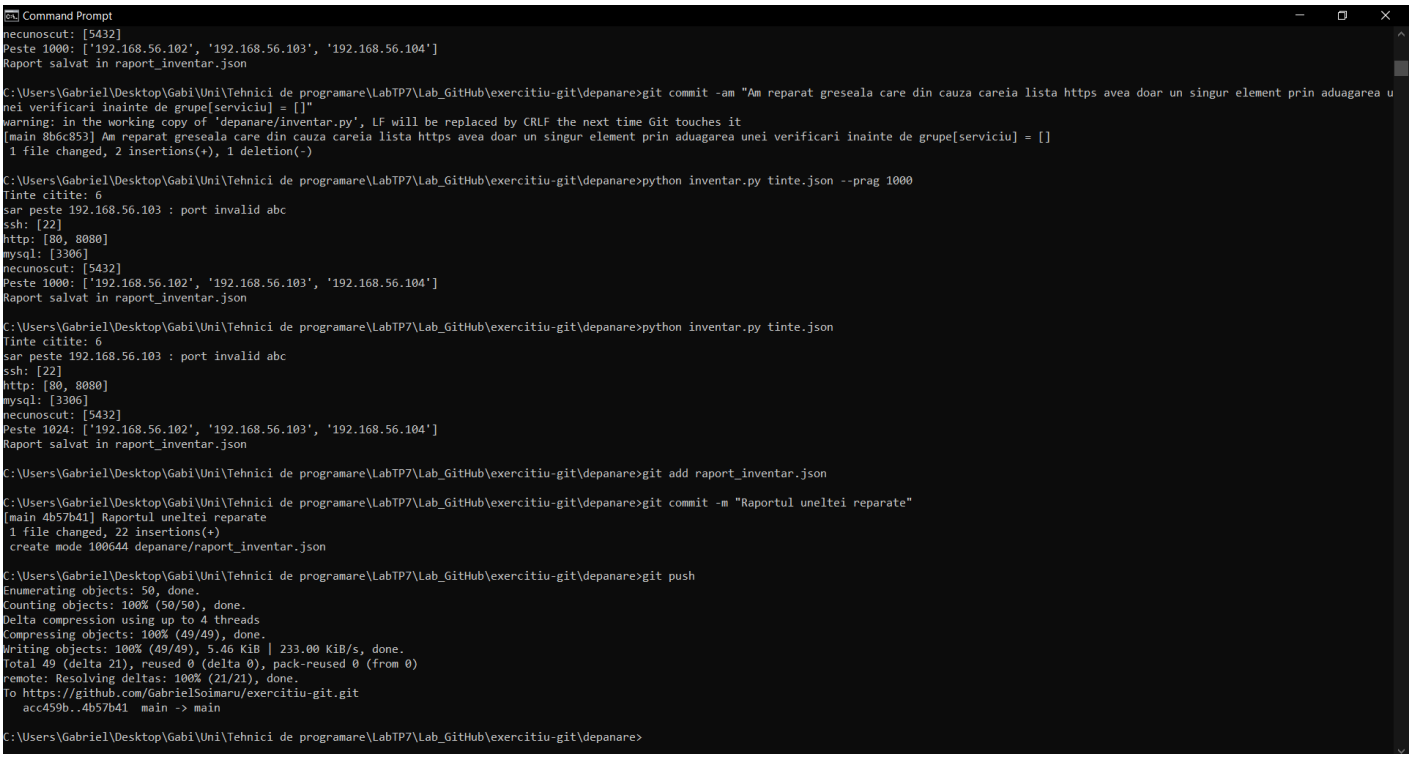


Figură 2.5.2 – Am tratat greșeala listei goale și am dat commit (B5)

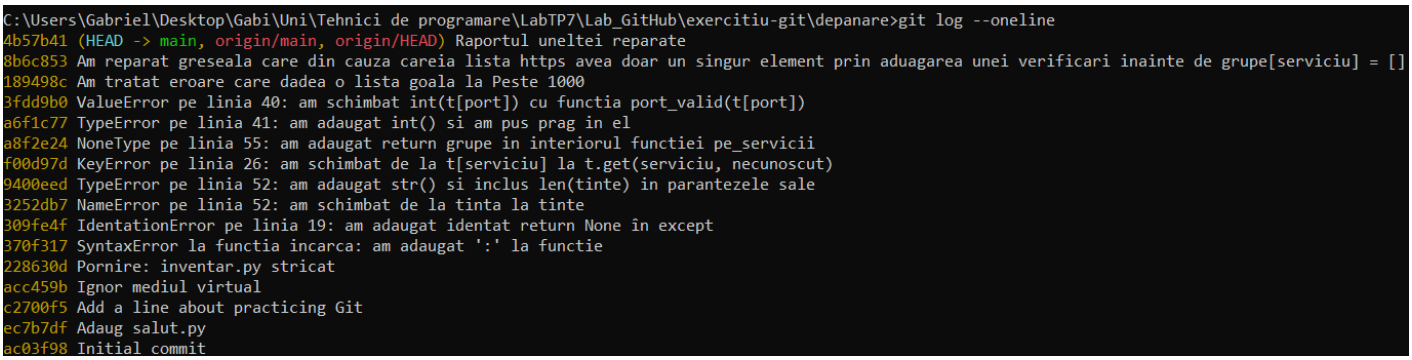


Figură 2.5.3 – Am tratat greșeala listei cu doar un element și am dat commit (B5)

B6



Figură 2.6.1 – Verificarea finală, adăugarea raport_inventar.json, commit și push pe git



Figură 2.6.2 – Toate cele reparațiile făcute

Link exertitiu-git: https://github.com/GabrielSoimaru/exercitiu-git.git

Link exertitiu-2: https://github.com/GabrielSoimaru/exercitiu-2.git

CONCLUZII

Pe parcursul acestei lucrări am învățat cum să lucrez cu GitHub și Git pe calculator. În partea A am creat mai multe depozite(repositories), am clonat de pe Git pe calculatorul personal și am dat push pe git. În partea B am modificat și reparat toate erorile din inventar.py făcând un commit la fiecare repare pentru a crea un istoric de reparații efectuate pe fișier, iar în final am dat push pe Git.

BIBLIOGRAFIE

1. Prezentare de Curs sesiunea 8 - https://else.fcim.utm.md/course/section.php?id=19133
2. Suport de Curs sesiunea 8 - https://else.fcim.utm.md/course/section.php?id=19133